



SPECIAL TOPIC ARTICLE

The metadata ecosystem and AI: Enabling FAIR and AI-ready data

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Abstract

Reproducibility is a foundational tenet of science. As artificial intelligence (AI) becomes increasingly embedded across science, the need to accurately document the provenance, structure, and behavior of training data, models, and workflows grows correspondingly. Metadata, understood as explicit and structured knowledge about data and related entities, is a critical yet often underexamined component of AI systems that helps address this need. High-quality metadata describing datasets, models, and workflows supports the FAIR (Findable, Accessible, Interoperable, Reusable) principles, strengthens reproducibility, and enables evaluation of AI-readiness by making data and models interpretable, traceable, and structurally consistent. Despite its central importance, sustained discussion of metadata as a core component of AI infrastructure remains limited. This article examines the role of metadata in advancing AI-enabled research, focusing on the design, implementation, and operationalization of metadata standards within an evolving metadata ecosystem. We first present a conceptual view of the metadata ecosystem, framed by data structure, data value, data encoding, and syntax standards, as a foundation for understanding how metadata enables AI and how AI contributes to metadata generation and refinement. We then introduce four case studies that illustrate how metadata can be generated, refined, and leveraged within AI workflows. The discussion synthesizes the cases, highlights limitations, including metadata quality challenges and the role of structured constraints in addressing AI errors, and relates each case to the metadata ecosystem dimensions it engages. Taken together, these cases illustrate

Abbreviations: AIC, Advance Information Collaboratory (University of Maryland, College Park); BGNN, Biology Guided Neural Networks; CAS, Computational archival science; ID4, Institute for Data-Driven Dynamical Design; KG, knowledge graph; MOF-KG, Metal Organic Frameworks-Knowledge Graph; LoS, Legacy of Slavery; MRC, Metadata Research Center (Drexel University); MOF, Metal Organic Framework; RAG, Retrieval-Augmented Generation.

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that metadata is not a peripheral add-on but an essential component of AI-ready, FAIR-aligned, transparent, and reproducible research systems.

INTRODUCTION

Artificial Intelligence (AI) is increasingly integrated into our daily lives, shaping how we gather information, communicate, and make decisions. Its influence ranges from routine activities, such as scheduling appointments and managing email, to complex processes including synthesis, modeling, reasoning, and prediction. Among its most significant impacts, AI has redefined and accelerated the research enterprise, transforming scientific and scholarly practices and procedures across nearly every discipline (Reddy and Shojaee 2025; Bianchi et al. 2025; Gridach et al. 2025). This shift is driven by the growing accessibility of low-barrier AI tools, together with the massive volume of data generated, collected, and used to train AI systems.

Metadata is a critical yet often overlooked factor in this AI evolution. Once chiefly viewed as a technical, practical application associated with library cataloging and database management, metadata has emerged as a vital component of AI, supporting many functions and serving as a valued data source (Musen et al. 2026; Del Ben 2025). Countless systems collect metadata as humans and machines interact in our highly networked, digital world—making phone calls, shopping online, and pursuing other basic tasks. This scale expands exponentially when considering systems with high-frequency sensors that collect data and metadata to support room lighting, temperature controls, or enable other monitoring. Adding to these sources is the volume of data and metadata produced by scientific instrumentation for example, microscopes, spectrometers, satellites. The expansive growth has positioned metadata as a foundational, indispensable building block of AI research. It supports data discovery, authenticity verification, provenance tracking, and other capabilities that embody the FAIR data principles—making data Findable, Accessible, Interoperable, and Reusable (FAIR) (Wilkinson et al. 2016), while also enabling reproducibility and ensuring data are AI-ready. Together, these three dimensions (FAIR, reproducibility, and AI-readiness) are increasingly referred to as FARR (FAIR in ML, AI-Readiness, and Reproducibility) (Huerta et al. 2023; Partners 2025; Kirkpatrick et al. 2022). The unique centrality of metadata in this evolving AI paradigm warrants closer analysis, as

we consider how AI will be developed and deployed in the future.

This article examines the relationship between metadata and AI. Specifically, we examine both directions of this relationship: metadata as infrastructure that enables AI, and AI as a mechanism for generating, refining, and managing metadata. The next section provides an overview of the metadata ecosystem as a foundation for this discussion, followed by several cases that demonstrate how AI can both generate and utilize metadata, reinforcing this bidirectional relationship. These cases draw on research conducted by the Metadata Research Center (MRC) at Drexel University and the Advanced Information Collaboratory (AI-Collaboratory) at the University of Maryland, College Park. A discussion follows, synthesizing the cases and identifying current limitations; and the conclusion considers the implications of metadata for AI. The overall aim of this work is to advance understanding of metadata in the context of AI, highlighting its challenges and potential as researchers seek to leverage this unique form of data as both infrastructure for AI and an output of AI-driven processes to support FAIR, reproducibility, and AI-readiness.

THE METADATA ECOSYSTEM

Every ecosystem has boundaries, though many are indistinct. A pond, for example, functions as an ecosystem, yet its boundaries shift with weather and atmospheric conditions. The metadata ecosystem is similarly diffuse, given the breadth of metadata-related applications and the ways both humans and machines interpret and use data and metadata. Despite the absence of a clear demarcation, there is general agreement about what metadata is and the landscape of standards that comprise the ecosystem.

Metadata defined

Metadata is explicit, encoded knowledge about an entity, expressed through declarative statements describing its properties, relationships, or contextual information. For example, metadata may record a dataset's creator or topicality, or an image's capture date. These statements function as assertions and are commonly encoded as property-value (or key-value) pairs, such as “creator = Smith” or

“date = 2026-01-21,” enabling exchange, validation, and computational processing.

In today’s computational ecosystem, metadata is generated for an expanding range of entities, from information-bearing objects for example, datasets, publications, recordings, to abstract constructs such as concepts and relationships. Metadata is typically organized into records–structured representations composed of one or more assertions—which may be expressed as property-value pairs or, at a finer level, as subject-predicate-object triples (Weibel 1995; Duval et al. 2002). FAIR-aligned frameworks similarly characterize metadata as machine-actionable collections of structured descriptions about identified digital objects (Jacobsen et al. 2020).

The wide range of entities and contexts to which metadata applies helps explain the familiar definition of metadata as “data about data.” More precise accounts describe metadata as structured information that supports functions, such as discovery, access, provenance tracking, relationship mapping (Greenberg 2010; Gilliland 2016; Greenberg 2017a). The functional aspect reflects intentional design choices and increasingly align with FAIR, reproducibility, and AI-readiness objectives. We next present a practical framework for examining metadata standards that reflect commonly recognized functional categories across domains.

The metadata landscape of standards

The metadata ecosystem comprises a broad landscape of interconnecting standards. The framework presented here organizes these standards into four high-level functional categories: *data structure standards*, *data value standards*, *data encoding standards*, and *data syntax standards*, as illustrated in Figure 1. This organizational framework draws from established treatments of metadata standards in information science (Greenberg 2017b; Zeng and Qin 2020). These categories often interconnect and may be packaged within a single standard. Each includes multiple examples—frequently referred to as schemas, systems, languages, or vocabularies—creating a diverse and occasionally confusing network. Reviewing these categories helps parse this landscape and clarifies how metadata enables AI while supporting FAIR, reproducibility, and AI-ready data.

Data structure standards

Data structure standards identify and define the descriptive properties that comprise a *metadata element set* for representing an entity. They support interoperability

by enabling systems to consistently interpret metadata assertions. A common form is the data dictionary used in database design. More formalized, machine-readable, shared data structure standards follow *ISO-11179 Information Technology – Metadata Registries (MDR)* (ISO/IEC 2015), which assign persistent identifiers (PIDs) to metadata properties. The Dublin Core (Dublin Core Usage Board 2022) is a well-known, general-purpose data structure standard in this category and is widely used to describe resources in digital libraries and repositories. More recently, data structure standards are emerging to address AI-related needs. Croissant (Akhtar et al. 2024), for example, was developed to support discovery, transfer, integration, and reusability of machine-learning datasets.

Data structure standards specify property requirements, indicate optionality, set cardinality, and establish constraints on values, among other rules. General-purpose standards such as Dublin Core operate primarily as flexible guidelines, allowing communities to define local requirements. In contrast, domain-specific data structure standards tend to be more prescriptive. Examples include the Crystallographic Information Framework (CIF) (Bernstein et al. 2016), which requires unit cell parameters, symmetry groups, and atomic positions; and the NetCDF Climate and Forecast (CF) conventions (Eaton et al. 2025), which mandate coordinate variables (time, latitude, longitude) and controlled vocabularies for naming units and variables.

Governance, specifically the development, maintenance, and sustainability of metadata element sets, is reflected in patterns of adoption and use. Some data structure standards emerge locally or institutionally, while others are stewarded by international research communities or formal standards bodies such as the International Organization for Standardization (ISO) or the National Information Standards Organization (NISO) in the United States. Broad adoption and use of data structure standards directly support FAIR by enabling discovery, interoperability, and reuse. When applied consistently, these standards further contribute to reproducibility and AI-readiness by enabling structured, machine-interpretable representations of scientific data.

Data value standards

Data value standards include controlled vocabularies, ontologies, taxonomies, name authority files, code systems, and related knowledge organization systems that constrain the assignment of values. They provide curated, endorsed sets of *values* (terms, codes, identifiers, or names) used to complete metadata property-value pairs. Standardizing permitted (controlled) data values supports

The Metadata Standards Landscape

- **Data structure standards** – identify and define descriptive properties (e.g., a data dictionary, or metadata element set)
- **Data communication standards (also known as encoding standards)** – specify metadata packaging and serialization mechanisms that support system operations and data transmission (e.g., XML, JSON)
- **Data value standards** – include name authority files, controlled vocabularies, ontologies, taxonomies, classificatory systems, and other mechanisms that constrain permissible values
- **Data syntax standards** – govern ordering, formatting, and syntactic rules that operate across data structure, communication, and value standards

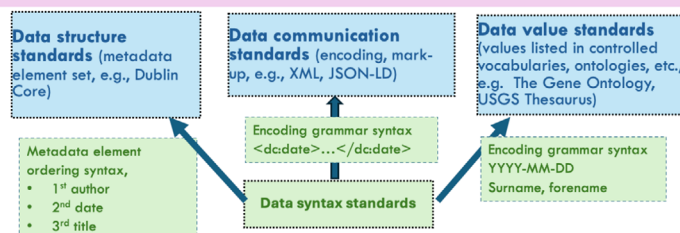


FIGURE 1 The metadata ecosystem.

semantic consistency, interoperability, and disambiguation across datasets.

Data value systems span general domains and broad subject areas, as well as highly specialized disciplinary contexts. One widely used general-domain value system is the *Library of Congress Subject Headings* (LCSH) (Library of Congress 2025), which supports topical description across many disciplines. Examples of more focused systems include the *Integrated Taxonomic Information System* (ITIS) (Guala 2019; National Museum of Natural History, Smithsonian Institution 2025) for species names and the *International Classification of Diseases* (ICD) (World Health Organization 2025), both of which serve multiple intersecting research and professional communities. More specialized value systems include the *Gene Ontology* (GO) (Consortium 2019) and the Periodic Table of Elements, each requiring substantial domain expertise to apply appropriately. These systems encompass terminological and classificatory structures used to standardize concepts, names, codes, measurements, and identifiers.

To support computational use, data value standards are typically made machine-readable through encoding standards and semantic models such as the Simple Knowledge Organization System (SKOS) (World Wide Web Consortium 2023b) and the Web Ontology Language (OWL) (World Wide Web Consortium 2009), as well as guidelines such as *NISO Z39.19: Guidelines for the Construction, Format, and Management of Monolingual Controlled Vocabularies* (National Information Standards Organization (NISO) 2019), discussed further below.

It is also important to recognize the role of provenance and administrative metadata, sometimes described as “metadata about metadata,” in supporting reproducibility and AI accountability. In many domains, ontologies

function simultaneously as value systems and structural frameworks, illustrating the fluid boundary between data structure and data value standards. Provenance models such as the W3C PROV Ontology (PROV-O) formalize authorship, temporal lineage, versioning, transformation histories, and life-cycle activities in machine-actionable form (Lebo et al. 2013; Missier et al. 2020). These standards enable transparent tracking of how data and metadata artifacts are created, modified, and reused, providing a foundation for accountable and reproducible AI-enabled research workflows.

Like data structure standards, value standards are governed by research communities and organizations and are commonly disseminated through registries that support discovery and reuse. For example, BioPortal (Vendetti et al. 2025), maintained by the National Center for Biomedical Ontology, provides access to hundreds of biomedical ontologies and related value systems. FAIRsharing (FAIRsharing Community 2025) similarly catalogs value standards alongside data structure standards. Although registries improve visibility, registered versions may not always reflect current releases, potentially exposing users to outdated or inconsistent content (Regino et al. 2021). Maintenance of currency has been shown to be challenging given the rapid evolution of scientific knowledge (Tudorache 2020; Ayerbe et al. 2022), although AI-assisted pipelines are increasingly being explored for ontology construction and maintenance (Matentzoglou et al. 2022; Fathallah et al. 2024; Kommineni et al. 2024). Overall, data value standards play a central role in shaping AI-ready datasets by supporting model training, enabling structured outputs such as knowledge graphs, and advancing FAIR data and reproducible research (Vogt et al. 2025; RDA FAIR Data Maturity Model Working Group 2020; Cox et al. 2021).

Data encoding standards

Data encoding standards specify how metadata *property-value pairs* or *subject-predicate-object triples* are formally represented, serialized, and packaged for exchange and computation. They provide structural rules that enable metadata to be transported, shared, ingested into pipelines, parsed for discovery, and processed across platforms. These standards support interoperability and preservation by offering common frameworks for representing metadata and underlying data models, including graph-based models such as the Resource Description Framework (RDF) (World Wide Web Consortium 2023a). Commonly used data encoding standards include eXtensible Markup Language (XML), JavaScript Object Notation for Linked Data (JSON-LD), and the MACHine Readable Cataloging (MARC) format widely used in libraries, with MARC records often rendered in XML (MARC-XML). Additionally, schema-driven modeling frameworks such as Linked Data Modeling Language (LinkML) (Saghaei et al. 2026) are increasingly used in bioinformatics, biomedical research, and data science to define machine-actionable data models that support FAIR and AI-ready datasets.

Scientific and scholarly encoding practices are frequently domain-specific. For example, NetCDF-CF encodes climate and geophysical data in self-describing binary files, while the Crystallographic Information Framework (CIF) represents crystallographic metadata as structured text. Data encoding standards play a critical role in AI pipelines by enabling reliable parsing and processing of large-scale datasets. They support the transformation of heterogeneous and unstructured research outputs into computationally accessible forms, underpin knowledge graph construction, and serve as essential enablers of FAIR data practices. Overall, data encoding standards facilitate machine interpretation, support reproducibility, and directly contribute to AI-ready data infrastructures.

Data syntax standards

Data syntax standards cut across data structure, value, and encoding standards by governing ordering, formatting, and syntactic conventions. Within data structure standards, syntax is reflected in the prescribed sequence of metadata properties. For example, the American Psychological Association (APA) citation style specifies the order *author, date, title*, while DataCite (DataCite 2025) requires the sequence *identifier (DOI), creator, title, publisher*. Domain-specific standards impose similar constraints: NetCDF Climate & Forecast (CF) mandates the order *time* → *latitude* → *lon-*

gitude, and the Crystallographic Information Framework (CIF) requires atomic positions to follow *atom label, atom type, x, y, z*. In each case, ordering conventions function as syntax rules that enable consistent interpretation.

Syntax also operates within data value standards, shaping how individual values are represented. For example, dates commonly follow the *year-month-day* format defined in *ISO 8601 Date and Time* (International Organization for Standardization 2019), which also specifies time values down to minutes and seconds. Personal names often follow conventions such as *surname, forename*, sometimes incorporating middle names, titles, suffixes, or degree designations. Similar syntactic patterns govern geographic names, species identifiers, disease codes, and other controlled values. Although implementation details vary across domains, the ordering and formatting of values directly affect how information is parsed, processed, and interpreted by computational systems, including large language models (LLMs).

Finally, syntax is an embedded component of data encoding standards, where it specifies delimiters, element placement, attribute structures, namespaces, and identifiers. Although user-facing software often hides these details, encoding syntax remains fundamental to machine readability. Together, data syntax standards support reliable parsing, enable interoperability across systems, and contribute to FAIR-aligned, reproducible, and AI-ready data infrastructures.

Metadata infrastructure and the metadata ecosystem

The standards categories reviewed above operationalize a data-intensive metadata ecosystem that increasingly underpins AI workflows. This ecosystem includes not only technical specifications but also a critical human dimension sustained through professional organizations, standards bodies, and disciplinary communities that collaboratively develop, revise, and maintain metadata standards through shared governance. It further encompasses resources that guide standards development and implementation, such as *ISO/IEC 11179: Information Technology – Metadata Registries (MDR)* (ISO/IEC 2015) and *ANSI/NISO Z39.19: Guidelines for the Construction, Format, and Management of Monolingual Controlled Vocabularies* (National Information Standards Organization (NISO) 2019), alongside semantic data models such as the Simple Knowledge Organization System (SKOS) (World Wide Web Consortium 2023b) and the Web Ontology Language (OWL) (World Wide Web Consortium 2009) noted earlier.

Together, these standards provide the infrastructure that makes metadata computable, interoperable, and reusable, supporting FAIR data and enabling machine learning pipelines and language models that depend on structured metadata as input. High-quality metadata, treated as a first-class data object, also serves as an invaluable AI data source by supporting discovery, model training, integration, and automated reasoning. This broader view helps explain the growing use of the term AI-ready data in research and industry. Although the term is not formally standardized, descriptions appear in industry guidance and community discussions (IBM 2024; Greenberg 2025). In this article, we define AI-ready data as data supported by structured, machine-actionable metadata that enables automated discovery, semantic alignment, provenance tracking, and reproducible computational use.

The metadata ecosystem reviewed in this section provides a practical framework for understanding the complexity of metadata and its role in supporting AI-ready data, as well as how AI methods, in turn, shape metadata through generation, validation, and refinement. Specifically, data structure, data value, data syntax, and data encoding standards are integral to rendering AI-ready data that are interpretable, computationally accessible, and capable of supporting AI workflows. The cases that follow illustrate how these components are operationalized in practice, highlighting metadata's varied roles in AI—from structuring inputs and governing vocabularies to enabling scalable, reproducible, and trustworthy workflows—and demonstrating both metadata as infrastructure for AI and AI as a mechanism for metadata generation and refinement.

AI METADATA-RELATED CASES

The massive scale of data and metadata, together with rapid advances in AI and computational technologies, has transformed research practices and procedures across nearly every discipline. These developments have further motivated the formation of new research communities with shared AI-driven goals and objectives. The cases presented in this article draw from two such communities: the NSF-supported Harnessing the Data Revolution (HDR) initiative (National Science Foundation 2021, 2025) and the Computational Archival Science (CAS) community (Marciano et al. 2018; Marciano 2022; Lemieux and Marciano 2025). One of the cases (AI-ready digital specimens) bridges both environments, illustrating the interplay between data science and evolving disciplinary frameworks in CAS.

Launched in 2018, the NSF Harnessing the Data Revolution (HDR) initiative was designed to help researchers

leverage the growing availability of big data, advances in computational power, and the expanding potential of AI. The HDR program began with seed grants, followed by the establishment of five national virtual HDR Institutes in the United States. The cases presented below draw from two HDR efforts: the Institute for Data-Driven Dynamical Design (ID4) and the Biology-Guided Neural Networks (BGNN) project, a seed grant that served as the foundation for the HDR Imageomics Institute.

In contrast, the CAS community emerged through a more grassroots process, as researchers began connecting archival practices with computational thinking in response to the growth of digital archives and big data. Since its launch in 2016, the CAS community has grown steadily, gaining international visibility through the annual IEEE Big Data Computational Archival Science (CAS) Workshop (Advanced Information Collaboratory (AICollaboratory) 2016; Lee et al. 2017), with December 2025 marking the tenth annual workshop. Together, these contexts set the stage for the four cases presented below, demonstrating how metadata standards and infrastructure are operationalized in AI-driven research environments and enable AI research.

Although the four cases arise from distinct disciplinary communities, each demonstrates reliance on specific components of the metadata ecosystem introduced earlier in Section 2. The AI-ready Digital Specimens and Synthesis Recipes cases emphasize the extension and refinement of data structure and data value standards within their respective domains. The MOF-KG and ChatLoS cases leverage data value standards, particularly ontologies, to support semantic precision, interoperability, and explainability. All four cases depend on data encoding and syntax standards that enable metadata to be represented, exchanged, and processed computationally. Taken together, these cases illustrate how the standards that comprise the metadata ecosystem contribute to AI-readiness in complementary and reinforcing ways. Table 1 summarizes the relationship between each case and the metadata ecosystem standards that facilitate contributions to FAIR and FARR.

AI-ready digital specimens

A key goal of the HDR initiative is for interdisciplinary teams to identify new ways to leverage existing data repositories for scientific discovery and innovation. This objective guided the HDR Biology-Guided Neural Networks (BGNN) project, which aimed to develop a novel class of artificial neural networks capable of exploiting machine-readable metadata and predictive knowledge associated with specimen images. The project initially focused on

TABLE 1 Mapping of case studies to primary metadata ecosystem standards.

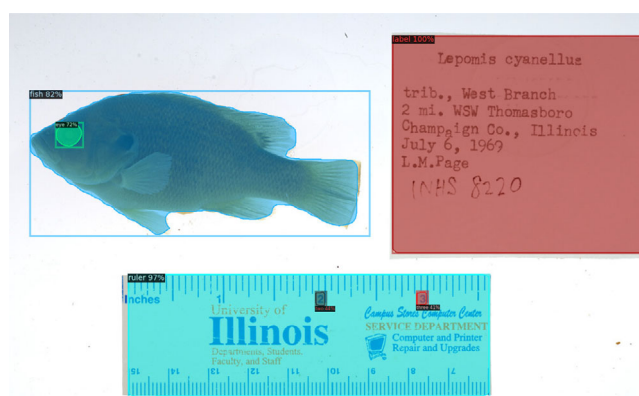
Case	Primary metadata focus	Metadata ecosystem standards
AI-ready digital specimens	Extension and validation of metadata properties aligned with existing schema for example, Darwin Core; correction of inconsistent values	Data structure, Data value
Synthesis recipes	Human-in-the-loop design of a domain-specific metadata schema for MOF synthesis procedures	Data structure
MOF-KG	Ontology development and semantic modeling to support knowledge graph construction and natural language querying	Data value
ChatLoS	Ontology-informed knowledge graph construction with provenance modeling and life-cycle relationships	Data value, data structure
All cases	Computational representation, exchange, and processing of metadata within AI workflows	Data encoding, syntax

images from the digital biodiversity repository iDigBio, motivated by the opportunity to advance scientific discovery using substantial existing investments in curated specimen collections (König et al. 2019).

In attempting to leverage fish specimen images for new scientific insights, the team found that metadata inconsistencies, omissions, and inaccuracies in major repositories were negatively impacting machine learning performance for species identification, segmentation, and related tasks by limiting the quality and interpretability of inputs to AI systems (Leipzig et al. 2020).

Addressing these metadata issues manually was time-intensive and infeasible at scale, motivating the need for automated and semi-automated approaches. As the project progressed, three core challenges became apparent: missing or insufficient metadata properties, incorrect or inconsistent metadata records, and the presence of poor-quality or inadmissible images that could not be systematically filtered.

To address these challenges, the project developed a suite of automated and semi-automated tools to generate, validate, and correct metadata, demonstrating how AI methods can enhance metadata quality at scale. These efforts support the broader goal of making collections AI-ready and FAIR. The first challenge was addressed by identifying a set of 38 core metadata properties (Leipzig et al. 2020) and developing methods to automatically generate these properties for each image (Pepper et al. 2021; Karnani et al. 2024). The tools employ AI-based methods (He et al. 2017; Wu et al. 2019) to identify and segment components within the fish specimen images. Figure 2 shows a fish specimen image that has been segmented as the first step in metadata generation. Once the components are isolated, conventional image and geometric techniques (Van der Walt et al. 2014) can be used to compute a variety of metadata for the specimen.

**FIGURE 2** Objects identified and segmented in a fish specimen image: fish, eye, ruler, '2', '3' and tag.

The second challenge was addressed by extracting textual information from specimen tags and comparing it against existing metadata records (Pepper et al. 2022). Optical character recognition and related AI-based techniques (Smith 2007; Smith et al. 2025) were used to identify and read tags embedded in images, with the extracted text evaluated for completeness, consistency, and accuracy (Navarro 2001). Custom GUI tools, as shown in Figure 3, were used to assess the quality of the computational results. Since image quality metrics are included in the 38 generated properties, they may be used to determine if a specific image qualifies for inclusion in a particular AI study. Inadmissible images may come in many forms: images of plants in a fish collection, radiographic versus photographic images (incorrect type), or a correct image type with an unacceptable layout structure. Developing generalizable methods for addressing these collection-level errors remains an open research challenge.

A major outcome of the BGNN project was the development of a fully integrated end-to-end workflow for collecting, preprocessing, segmenting, and analyzing specimen

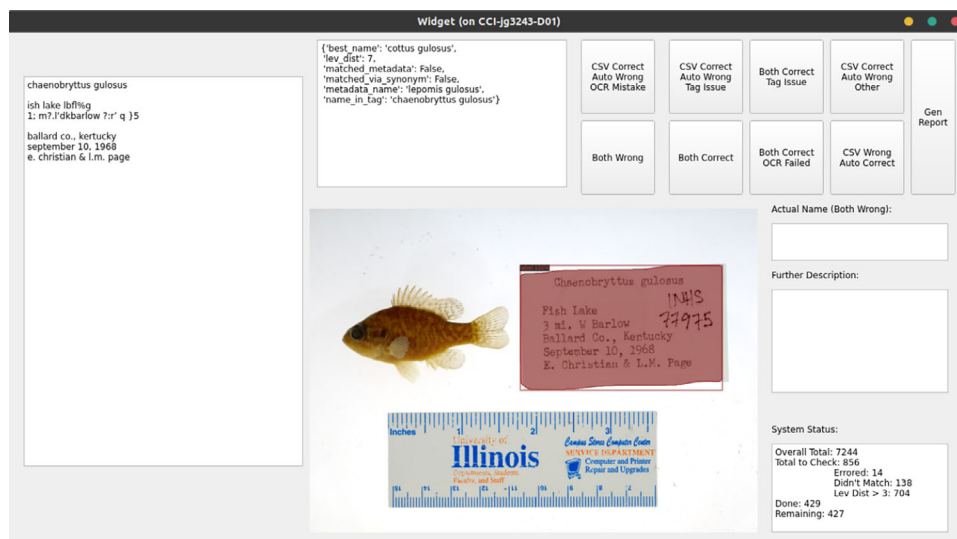


FIGURE 3 Graphical user interface for assessing tag analysis results.

image collections at scale. The AI-based automated metadata generation methods detailed here play an important role in enabling downstream AI workflows by providing structured, machine-interpretable inputs (Balk et al. 2024). In a follow-on study, information about the outline, shape, and size of fish was used as a proof of concept to demonstrate the usefulness of metadata within an ML application (Breen et al. 2023). This study developed a species identification technique that relied exclusively on mathematical representations of fish morphology automatically derived from specimen images. Overall, this case illustrates how high-quality, systematically generated metadata not only supports FAIR principles but also becomes a first-class computational asset in AI-driven scientific workflows. It further demonstrates the bidirectional relationship between metadata and AI, specifically that structured metadata enables improved machine learning performance, while AI methods generate, validate, and enhance metadata at scale.

Synthesis recipes: Designing materials synthesis metadata

Scientific papers are a major source of cutting-edge knowledge in materials science, containing valuable information such as experimental results, material properties, and chemical structure details (Weston et al. 2019; Tshitoyan et al. 2019). These data play an important role in facilitating materials discovery with data-driven ML methods (Kim et al. 2017; Park et al. 2022). However, the sheer volume and rapid growth of research publications create a major bottleneck for researchers. Manually searching and reading relevant papers is time-consuming; the peer-reviewed

literature already numbers in the millions, and its exponential growth makes this task increasingly unmanageable (Glasby et al. 2023; Zhao et al. 2021).

For Metal-Organic Frameworks (MOFs), a type of material with high potential in multiple fields such as medicine and environmental science (Furukawa et al. 2013; Yaghi et al. 2003; Glover and Mu 2018; Lee et al. 2009), the need for automated knowledge extraction is particularly pressing (Zheng et al. 2023; Luo et al. 2022). The synthesis procedures of MOF materials remain largely empirical (Lee et al. 2021), with much of the practical knowledge buried in unstructured descriptions spread across tens of thousands of research articles. In addition, the reporting of synthesis procedures can vary in the level of detail and style. To address these reporting disparities, a well-defined synthesis metadata schema is essential because it specifies how synthesis procedures can be recorded without significant information loss (Greenberg et al. 2023). For example, it defines which information researchers seek and how it should be represented. This serves as a blueprint for the actual synthesis extraction task. Translating textual descriptions of experimental details into fully structured, computationally-ready records based on such metadata can support material computational experiments with a vast number of samples, enable predictive modeling and synthesis optimization, and ultimately accelerate MOF discovery (Park et al. 2022; Glasby et al. 2023).

To enable automatic extraction of MOF synthesis experiment data across a large volume of literature, our approach comprises two main components. First, we designed a synthesis metadata schema through a human-in-the-loop process, in which MOF domain experts and computer scientists iteratively defined and refined entities, inter-entity relationships, and the sequential steps required to

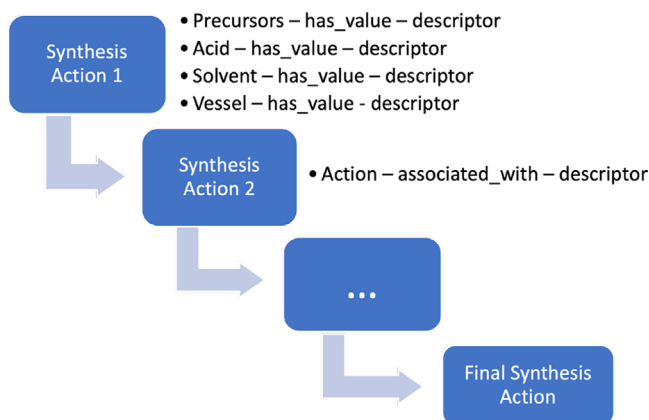


FIGURE 4 Metadata properties for the MOF synthesis.

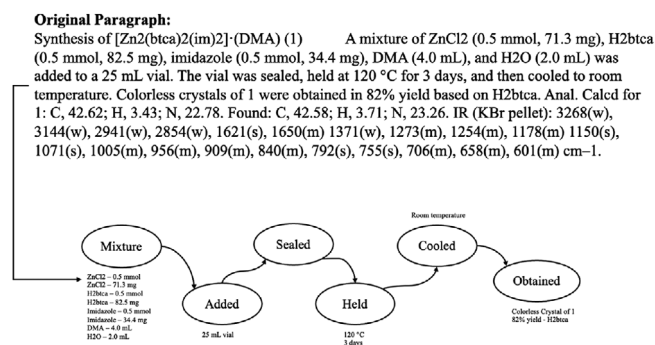


FIGURE 5 An example of codified synthesis recipe extracted from research paragraph (Example paragraph from (Liu et al. 2014)).

represent synthesis procedures (Figure 4), detailed in (Zhao et al. 2023). The schema is represented as a directed graph encoding the sequential structure of synthesis actions, with each node corresponding to a synthesis step and its associated metadata attributes. This metadata schema also serves as a guide for downstream LLM-based synthesis extraction (Zhao et al. 2025).

Second, MOF-related research papers were matched to MOF Cambridge Structural Database (CSD) entries (Groom et al. 2016) and converted from XML to text. MOF synthesis paragraphs were identified using LLM-based text classification. Once identified, we implemented a prompt-based, few-shot named entity and relation extraction pipeline using the OpenAI API for GPT-4o. A small set of expert-annotated synthesis procedures was provided as examples, enabling the model to codify each synthesis procedure into a machine-readable graph representing synthesis steps and their attributes. The machine-readable graph representation is shown Figure 5, and the MOF CSD workflow is presented in Figure 6.

Overall, the synthesis paragraph classification model achieved an accuracy of 0.94 and a macro-averaged F1 score of 0.93 (Zhao et al. 2025). Using this approach, we produced 3,583 fully structured synthesis records, enabling

large-scale analysis of MOF synthesis experiments through a structured linear graph framed by metadata properties.

Metal organic frameworks—knowledge graph (MOF-KG)

Materials science discovery requires sifting through vast amounts of data for targeted applications. Traditionally, this process is slow and time-consuming because relevant knowledge is scattered across scholarly articles, laboratory notes, and specialized databases. By consolidating this knowledge in a metadata-rich knowledge graph, domain scientists can rapidly search for relevant information, accelerating the discovery cycle. This work uses Metal-Organic Frameworks (MOFs) as a case study to demonstrate the advantages of knowledge-graph-supported question answering using natural language (An et al. 2023). MOFs hold substantial potential for applications such as gas storage, separation, catalysis, and drug delivery, yet critical information about their structures, properties, and synthesis procedures remains distributed across databases and literature. To address this fragmentation, we integrated these dispersed data sources into a knowledge graph (MOF-KG) that subscribes to a domain-specific MOF ontology (Figure 7).

The AI component of this work centers on a large language model (LLM)-powered natural language query interface. This interface enables domain scientists to query the knowledge graph without requiring formal training in query languages such as Cypher or SPARQL. Beyond mapping natural language to formal graph queries, the system addresses challenges related to domain-specific terminology, ambiguous phrasing, and complex query structures. The solution is built on a strong metadata foundation where each concept, entity, relationship, and property is semantically enriched with identifiers defined in the MOF ontology, significantly improving semantic mapping, interoperability, and query accuracy.

To develop and evaluate the LLM-powered natural language query engine, we compiled a benchmark dataset containing 161 complex MOF-related natural language questions, each rephrased in multiple variants, resulting in 644 total questions. This dataset was used to train and evaluate the query engine. For training, each question was paired with a corresponding Cypher query annotated with chain-of-thought (CoT) reasoning steps. To translate natural language questions into formal queries, we tested six methods ranging from zero-shot ontology-only prompting to few-shot learning (Wang et al. 2020) enriched with ontology definitions and CoT reasoning. Embedding-based semantic similarity was used to match test questions to relevant training examples for few-shot configurations.

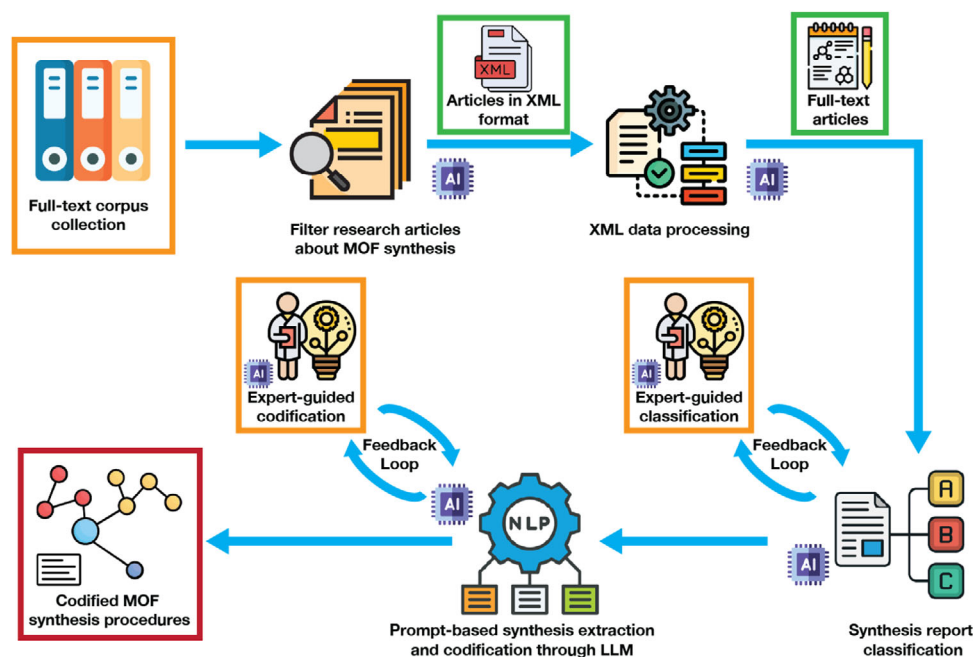


FIGURE 6 LLM-enhanced framework for MOF synthesis metadata codification.

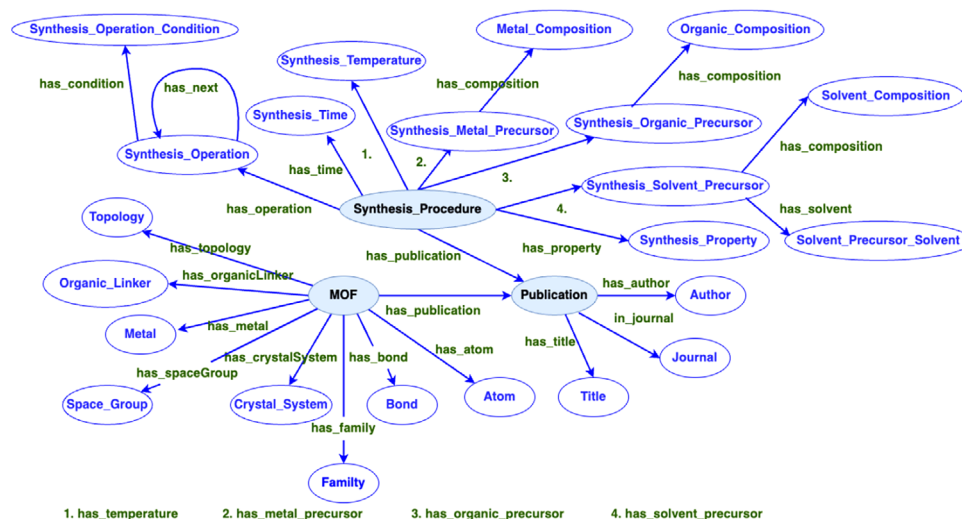


FIGURE 7 MOF knowledge graph.

Results indicate that combining example question-answer pairs, ontology definitions, and CoT reasoning achieved the highest performance, with an F1-score of 0.891. Table 2 summarizes the evaluation outcomes.

This work demonstrates that coupling AI tools such as LLMs with metadata resources such as ontologies can substantially improve access to complex domain knowledge. In the MOF-KG project, we developed a MOF ontology designed to make MOF data findable and accessible, and constructed a knowledge graph subscribing to this ontology (An et al. 2022, 2022), enabling inter-

TABLE 2 Performance of various methods for the NL query engine.

Method	F1-score
Zero-shot ontology only	0.411
1-shot Q&A pair	0.829
1-shot Q&A pair + ogy	0.845
1-shot Q&A pair + CoT reasoning	0.876
1-shot Q&A + ontology + CoT reasoning	0.891

operability across heterogeneous MOF data sources and directly supporting FAIR principles. Given the ontology as a metadata foundation, the LLM was able to translate natural language questions into formal knowledge graph queries with high accuracy, enabling efficient retrieval of targeted results. Both the MOF-KG and the benchmark dataset are published in open repositories, supporting reuse and continued development. Together, this work illustrates how metadata-enriched knowledge graphs can bridge domain expertise and computational querying in AI-driven scientific discovery.

Enhancing access to digital archival records using generative AI

This section presents work from the Computational Archival Science (CAS) field by the Advanced Information Collaboratory (AIC). It expands upon AIC's prior implementation (Gnanasekaran and Marciano 2024) of a conversational research assistant, "ChatLoS," which integrates Knowledge Graphs (KGs) with Retrieval-Augmented Generation (RAG) to improve access to Maryland's Legacy of Slavery (LoS) digitally transcribed archival collections. The project demonstrates how metadata is integral to AI by transforming archival description into a computable knowledge graph that grounds large language model (LLM) responses. In this broader archival context, cultural heritage collections are often rich but siloed and item-centric, making it hard to trace people and events across record types or answer multi-hop questions. KGs can surface relationships among items and support narrative synthesis, while RAG grounds large language model outputs on retrieved evidence to improve factuality. This contribution places archival metadata at the center of RAG. Instead of vector-searching text, retrieval occurs from a Neo4j KG via Cypher queries, enabling the LLM to explain results in natural language, thereby providing trustworthy results.

In RAG, LLMs are grounded by evidence fetched at runtime. When the retrieval source is a KG, models can draw on authoritative, curated relationships and provenance, which improves factuality and explainability compared to text-only retrieval (Lewis et al. 2020). Together, metadata and KGs make collections AI-ready by supporting precise retrieval, transparent reasoning paths, and durable integration. This graph structure directly benefits AI systems: it provides a semantic backbone over which retrieval and reasoning can operate. Metadata is the connective tissue that makes data computable for AI by enabling well-described entities, relationships, identifiers, and provenance that machines can reliably find, integrate, and reuse. The FAIR principles formalize these requirements and

have shaped international policy for data stewardship (Wilkinson et al. 2016).

The AIC implemented a metadata-first KG-RAG pipeline that turns archival description into computable structures and explainable answers. Digitally archived collection records from the LoS datasets (certificates of freedom, manumissions and domestic traffic ads) were cleaned, normalized, and mapped to an application ontology; entities such as people, locations for example, counties, and documents receive persistent identifiers, and evidence-level provenance retains page images and catalog references. The resulting KG is stored in Neo4j with nodes and relationships, to support fast multi-hop traversals and aggregations. On top of the KG data store, a Graph-RAG layer converts natural-language questions into parameterized Cypher using constrained prompt templates and a small toolkit of query schemata (paths, counts, filters, time windows). Queries execute in the database and return compact graph/tabular payloads with source identifiers. A QA layer then composes the final response from these payloads, surfacing citations, rendering links or snippets, and explaining the reasoning path. Guardrails such as "answer-only-from-graph" prompting, and visible execution logs reduce hallucination and improve explainability. This design makes archival metadata AI-ready end-to-end, enabling reproducible narratives, aggregate statistics, and provenance-aware multi-hop tracing across collections.

The system architecture consists of three coordinated layers:

1. Knowledge Graph Layer: Models archival entities and relationships in Neo4j, including multiple LoS collections such as Domestic Traffic Ads, Manumissions, and Certificates of Freedom.
2. Graph-RAG Layer: Translates natural language questions into Cypher queries to retrieve structured data from the KG.
3. LLM QA Layer: Uses a LLM to compose final answers grounded in the Cypher query results, improving factual accuracy.

In Table 3 we see the node labels for example, Enslaved-Person, SlaveOwner, DomesticTrafficAd, Manumission-Record, and corresponding typed relationships that form the KG created using the Neo4j database. The ChatLoS powered by KG-RAG reliably reconstructs person-level trajectories across archival sources, linking domestic traffic advertisements, manumission records, and certificates of freedom into cohesive narrative answering directly from the graph. In a representative workflow, a natural-language question is translated into Cypher, executed against the KG to retrieve a precise piece of data, and the

TABLE 3 ChatLoS Neo4j KG–Core nodes and representative properties.

Node label	Representative properties (examples)
EnslavedPerson	first_name, last_name, sex, age, complexion, alias
SlaveOwner	full_name, owner_first_name, owner_last_name
DomesticTrafficAd	sale_date, newspaper_code, county, skills_specified
ManumissionRecord	date_manumitted, date_recorded, county_name, witness_fullname
CertificateOfFreedom	date_freed, series, item, page, Freed_FirstName, Freed_LastName

returned evidence is then verbalized as a compact, source-anchored explanation. This end-to-end path produces answers that are simultaneously narrative and explainable.

In the ChatLoS system, metadata functions as a first-class computational layer that directly enables retrieval, reasoning, and explanation by the AI. Archival metadata extracted from Domestic Traffic Ads, Manumission Records, and Certificates of Freedom is ingested as structured node properties in Neo4j, including personal identifiers for example, `first_name`, `last_name`, `age`, `sex`, and temporal fields for example, `sale_date`, `date_manumitted`, `date_freed`, and geographic descriptors for example, `county`, `county_name`, and record-level provenance for example, `newspaper_code`, `series`, `page`, `ad_image_metadata`. These properties are not embedded in narrative text but are normalized into discrete, queryable fields, allowing the system to align records across datasets and over time. Explicitly typed relationships such as `LISTED_FOR_DOMESTIC_TRAFFIC_SALE_IN`, `MANUMITTED_THROUGH`, `WAS_GRANTED_FREEDOM_THROUGH` encode legally and historically meaningful events, enabling deterministic multi-hop traversal between individuals, transactions, and documentary evidence. The resulting knowledge graph materializes this metadata and provides an executable structure over which retrieval and reasoning operate.

Computable metadata is essential to the system's AI behavior in ways that free-text archival description alone cannot support. Tracing an individual's trajectory from sale to manumission and certification of freedom requires explicit temporal ordering, stable identifiers, and semantically constrained relationships that cannot be inferred reliably from prose. For example, the system's ability to retrieve "`sale_date`" attribute from a Domestic Traffic Ad, align it with "`date_manumitted`" from a Manumission Record, and confirm recognition through "`date_freed`" attribute in a Certificate of Freedom depends on these fields being modeled as structured metadata rather than narrative text. Natural-language questions are translated into Cypher queries, as shown in Figure 8, that operate exclusively over these properties and relationships, and return compact, provenance-aware results that include links to original newspaper images, archival series, and page references. The language model is restricted to

composing responses from this retrieved evidence, using metadata-encoded sources to explain both outcomes and reasoning paths. In this way, metadata does not merely support FAIR discovery and interoperability, but also actively constrains AI generation, reduces hallucination, and renders historically sensitive archival collections AI-ready for transparent and reproducible narrative synthesis.

Figure 9 shows how ChatLoS links summaries and graph traversals back to original document images. Including a scanned newspaper domestic traffic ad stored in MSA's online database as evidence shows how the graph's entities and relationships map to real cataloged items and page images. It also underscores the system's commitment to provenance and responsible presentation of historically sensitive language contained in primary sources.

In Figure 10, the KG-RAG trace ("Tracing the Life") exposes provenance links primarily to Domestic Traffic Advertisement (DTA) records, while corresponding links to Manumission and Certificate of Freedom records are not shown. This reflects a current limitation of the underlying knowledge graph rather than an absence of conceptual support for multi-source provenance. At present, DTA records are represented with stable identifiers and resolvable archival references, enabling direct link-backs to digitized sources (as illustrated in Figure 9). In contrast, Manumission and Certificate of Freedom records are modeled at the entity and relationship level, but lack uniformly defined graph attributes for example, persistent URLs, archival identifiers, required to generate consistent, user-facing provenance links. Addressing this limitation will require extending the graph schema to include source-specific provenance attributes, followed by re-ingestion of affected collections and regeneration of relevant KG queries. Additionally, public online availability of Manumission and Certificate of Freedom records varies across archival systems, which currently constrains the feasibility of direct link-back presentation in all cases. As a result, Figures 8, 9, and 10 should be interpreted as illustrating partial but evolving provenance support, with DTA records serving as the most fully instrumented example at this stage. Improving uniform provenance exposure across heterogeneous historical datasets remains an active area of development.

This progression strongly suggests that John Howard transitioned from being enslaved, being listed for sale, gaining his freedom, and then receiving official recognition of that freedom. This kind of detailed tracing can further our understanding of the process of liberation during this period.

● Generative AI explanation behind the response

What the AI did:

The AI interpreted your question: 'Can you trace the journey of individuals from being listed in Domestic Traffic Ads to receiving a Certificate of Freedom, including any manumission records?', generated a Cypher query to explore graph relationships in the Knowledge Graph, executed it against Neo4j, and composed an answer from the query results.

Internal Logs (Advanced):

```
> Entering new GraphCypherQChain chain...
Generated Cypher:
cypher
MATCH (e:EnslavedPerson)-[:LISTED_FOR_DOMESTIC_TRAFFIC_SALE_IN]->(ad:DomesticTrafficAd)
(e)-[:MANUMITTED_THROUGH]->(m:ManumissionRecord),
(e)-[:WAS_GRANTED_FREEDOM_THROUGH]->(f:CertificateOfFreedom)
RETURN e.first_name AS FirstName,
       e.last_name AS LastName,
       ad.Ad_Image_Metadata AS AdImageLink,
       ad.Sale_Date AS SaleDate,
       m.DateManumitted AS ManumissionDate,
       f.date_freed AS FreedomDate
ORDER BY FirstName, LastName, SaleDate
```

Full Context:

```
[{'FirstName': 'John', 'LastName': 'Howard', 'AdImageLink': 'http://msa
```

```
> Finished chain.
DEBUG: Chain Result: {'query': 'Can you trace the journey of individual
```

 Clear Chat

FIGURE 8 Example of ChatLoS powered by KG-RAG-Cypher query.

The ChatLoS results support the FAIR principles in practice in multiple ways. The KG makes entities and relationships Findable and Accessible via natural-language interaction. The consistent schema and use of persistent identifiers support Interoperability across collections; and the queries, subgraphs, and identifiers render results Reusable. Together, the combination of well-modeled metadata, constrained query generation, and explicit provenance delivers AI-ready outputs that are transparent and trustworthy. These aspects help to generate results that are both insightful for users and verifiable for researchers, advancing provenance-aware discovery across cultural-heritage collections. In addition, the structured retrieval

and natural-language explanation support expert and public exploration of cultural archives, enhancing access to historically sensitive collections while advancing FAIR and FARR principles in the digital heritage domain. Overall, the ChatLoS project demonstrates how metadata modeled in a knowledge graph can serve as the foundation for an AI-powered discovery system.

DISCUSSION

The cases presented in this article underscore the central role that metadata plays in enabling AI research



FIGURE 9 ChatLoS provided link to the source document stored on MSA's online database.

ChatLoS enhanced with KG-RAG using Neo4j and Generative AI—Cross-dataset querying with DTA, Manumissions, and Certificates of Freedom

Can you trace the journey of individuals from being listed in Domestic Traffic Ads to receiving a Certificate of Freedom, including any manumission records?

Yes, based on the available data, we can trace the journey of an individual named John Howard from being listed in Domestic Traffic Ads to receiving his Certificate of Freedom.

- Domestic Traffic Ads:** John Howard was listed for sale in a Domestic Traffic Ad on August 15, 1829. You can view the original ad through the following source: 1829 Domestic Traffic Ad for John Howard.
- Manumission Records:** Approximately a year later, his Manumission Record shows that he was granted freedom on September 28, 1830.
- Certificate of Freedom:** Subsequently, he was officially recognized as a free individual through his Certificate of Freedom dated October 25, 1830.

This progression strongly suggests that John Howard transitioned from being enslaved, being listed for sale, gaining his freedom, and then receiving official recognition of that freedom. This kind of detailed tracing can further our understanding of the process of liberation during this period.

Generative AI explanation behind the response

What the AI did:

The AI interpreted your question: "Can you trace the journey of individuals from being listed in Domestic Traffic Ads to receiving a Certificate of Freedom, including any manumission records?". Generated a Cypher query to explore graph relationships in the Knowledge Graph, executed it against Neo4j, and composed an answer from the query results.

FIGURE 10 Example of ChatLoS powered by KG-RAG.

across diverse applied contexts, while also highlighting distinct methodological approaches. Across these cases, the relationship between metadata and AI is broadly bidirectional, with metadata providing structured inputs that enable AI systems, and AI contributing to the generation, refinement, and validation of metadata.

The first two cases (Section 3.1 AI-ready Digital Specimens and Section 3.2 Synthesis Recipes) address research bottlenecks through the lens of data structure and data value standards. The AI-ready digital specimens case implemented automated metadata generation and validation to remediate metadata challenges that were impeding AI workflows. These challenges included incomplete and inconsistent implementation of data structure standards for example, Darwin Core, and poorly controlled metadata values. In this context, metadata functions both as a gatekeeper by confirming whether specimen images are suitable for AI analysis, and as a value-added layer that extends and corrects existing records (Karnani et al. 2024; Leipzig et al. 2020). By contrast, the Synthesis Recipes case addressed the overwhelming volume of unstructured textual data in materials science publications. Domain experts and computer scientists collaboratively defined key MOF synthesis entities and relationships, effectively establishing a base-level data structure standard in the form of a metadata schema. LLM-based extraction pipelines were then applied to codify experimental synthesis procedures, transforming narrative descriptions into computable representations (Zhao et al. 2025). Together, these two cases illustrate how data structure and data value standards can be extended, refined, and operationalized to produce AI-ready data support FAIR and FARR principles.

The knowledge graph-driven cases (Section 3.3 MOF-KG and Section 3.4 ChatLoS) further demonstrate how metadata enables AI systems to retrieve, reason over, and explain results (Lewis et al. 2020). Both projects leverage data value standards through ontologies that provide semantically precise definitions of entities, properties, and relationships. In MOF-KG, ontology development supported the construction of a metadata-rich knowledge graph designed for scalable, domain-specific discovery via natural language queries (Khan 2023; Wang et al. 2024). In ChatLoS, archival description was transformed into structured, provenance-aware knowledge graphs that ground RAG, enabling multi-hop tracing of entities and historically meaningful events across record types. In this context, provenance modeling functions as both a data structure and data value standard, encoding life-cycle relationships that enhance transparency and trust. Although both projects employ metadata-driven knowledge graphs, MOF-KG emphasized scalability and semantic enrichment for scientific discovery, whereas ChatLoS emphasized explainability, provenance, and responsible presentation in cultural heritage contexts.

These cases collectively illustrate that metadata plays a foundational role in shaping AI-driven research workflows, while also revealing limitations that require careful attention. Metadata infrastructure depends on interrelated

and sometimes competing standards. *Data structure standards*, *data value standards*, and *data encoding standards* require ongoing revision and sustained development, which in turn influences *syntax standards* and long-term interoperability. Quality deficiencies can directly degrade AI performance, as demonstrated in the specimen image work, where incomplete or inconsistent metadata reduced model accuracy. Similarly, inconsistent reporting of synthesis methods complicates large-scale extraction and structured codification.

Across the knowledge-graph-based cases, metadata-driven retrieval and constrained query execution play a critical role in reducing hallucinations by grounding AI outputs in explicitly modeled entities, relationships, and provenance. Rather than allowing unconstrained generation, systems such as MOF-KG and ChatLoS restrict responses to structured graph queries and retrieved evidence. Although quantitative hallucination rates were not systematically captured across all four cases, each project incorporated schema constraints, provenance checks, and human validation loops to identify and correct faulty outputs. These safeguards reflect an emerging pattern; specifically that high-quality, computable metadata does not eliminate AI errors, although it significantly constrains them and facilitates detection and auditing (Lewis et al. 2020).

At the same time, knowledge graph projects require substantial computational investment to maintain semantic alignment, trustworthiness, and continued operation. Human-in-the-loop activities remain essential, particularly during schema design, extraction validation, and provenance verification, even as computational approaches improve scalability. Evaluating metadata quality and developing standardized mitigation strategies are therefore critical, and increasingly urgent as data and metadata continue to expand toward entropy (Habermann 2020). Collectively, these cases demonstrate that metadata is not merely supportive infrastructure, but a governing layer that shapes how AI systems retrieve, interpret, and justify knowledge. Although metadata enables FAIR, AI-ready, and reproducible research workflows, sustaining and scaling the broader ecosystem of standards remains an ongoing technical and institutional challenge.

CONCLUSION

This article examined the role of metadata in AI by defining key foundations, introducing the metadata ecosystem framework, and presenting four cases grounded in AI-metadata applications. These cases address AI-ready digital specimens and metadata quality, metadata for

structuring synthesis protocols and advancing materials discovery, and the use of knowledge graphs to enable large-scale integration and reasoning across diverse data sources. The discussion highlights both the strengths and limitations of current metadata practices. Taken together, the cases demonstrate that metadata is not a peripheral add-on, but a central and enabling component of AI-driven research workflows.

It is important to underscore that the cases presented here represent only a small portion of a rapidly expanding research area. More importantly, they position metadata as a substantive and necessary area of inquiry in its own right. Advancing AI requires sustained attention to how standards are designed, implemented, evaluated, and sustained. Sharing methods, protocols, successes, and limitations across communities is essential to strengthening practice and expanding opportunities in AI contexts. The work described across these cases aligns with ongoing efforts to support FAIR principles, reproducibility, and AI-ready data, and connects to broader conversations about transparency and trustworthiness in AI systems reflected in FARR. Importantly, this work highlights the bidirectional nature of this relationship, where metadata serves as essential infrastructure for AI, while AI increasingly contributes to metadata generation, enhancement, and management.

Looking ahead, metadata will continue to shape how AI systems are designed, evaluated, and governed. As AI systems become increasingly embedded in scientific, cultural, and civic infrastructures, the standards that comprise the metadata ecosystem will influence what can be discovered, verified, and reproduced. Sustaining this ecosystem requires ongoing technical refinement, institutional stewardship, and cross-disciplinary collaboration. The future of AI will not be written in code alone, but through the structured, governed, and interoperable metadata that makes knowledge findable, usable, and accountable.

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